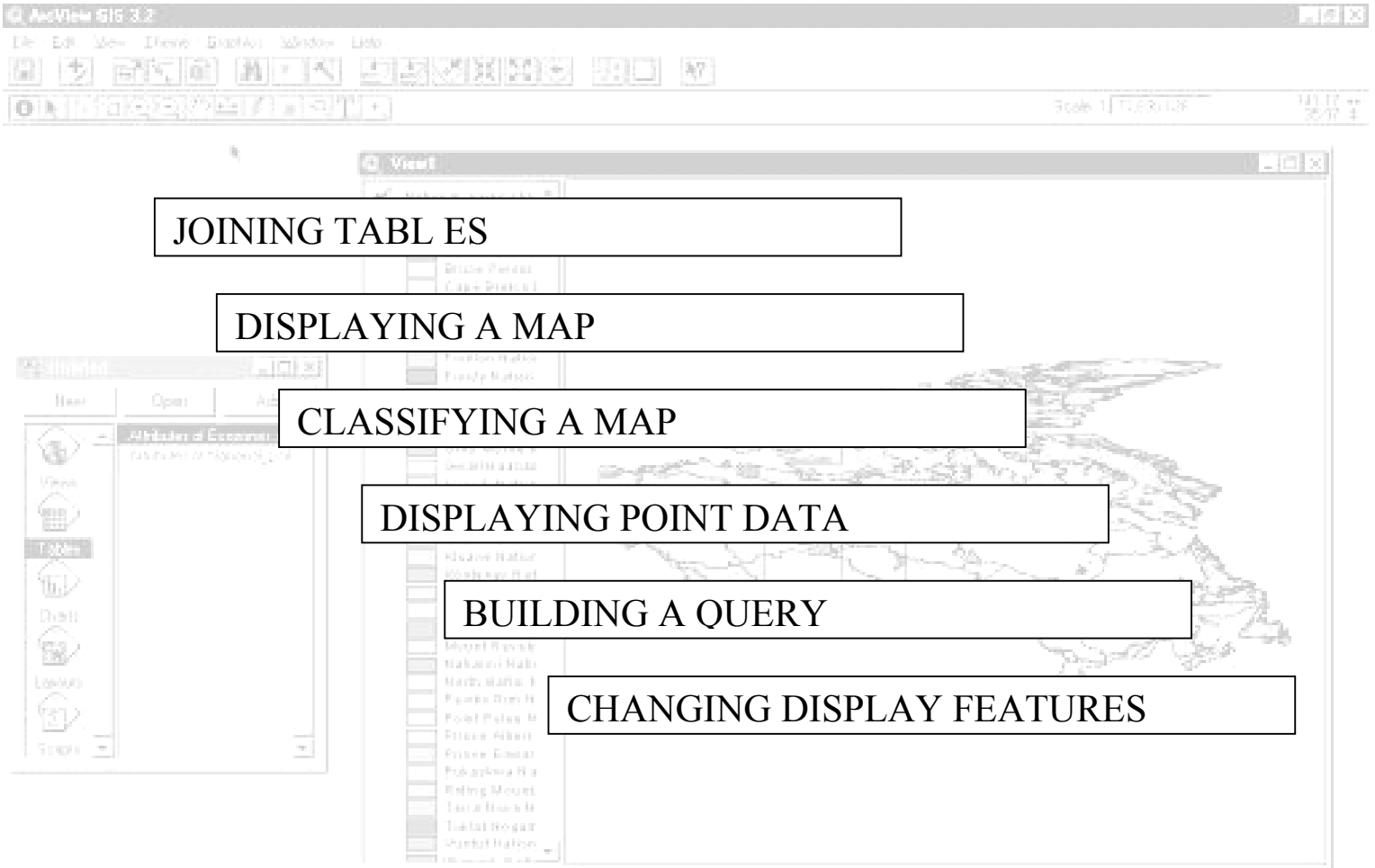


Geological Diversity and Geological History of Canada



GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

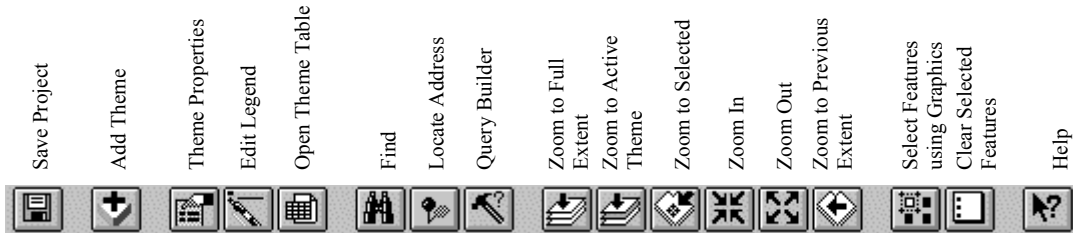
Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

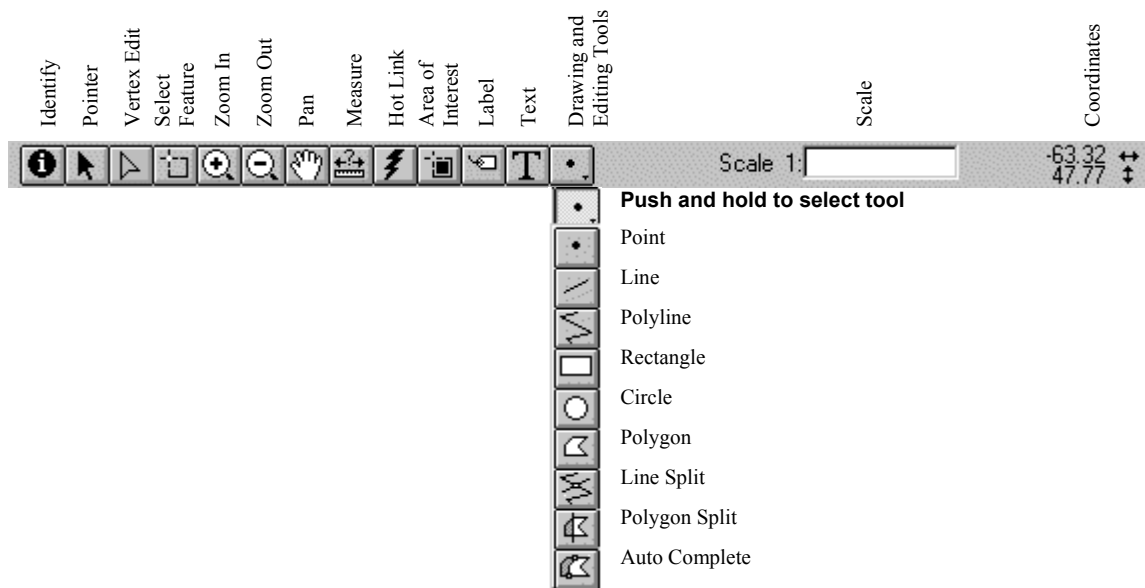


Table - Buttons



Table - Tools



Chart - Buttons

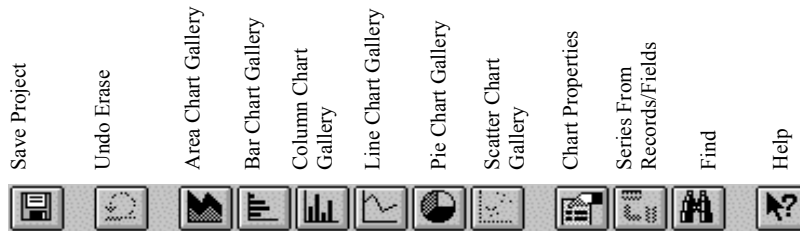
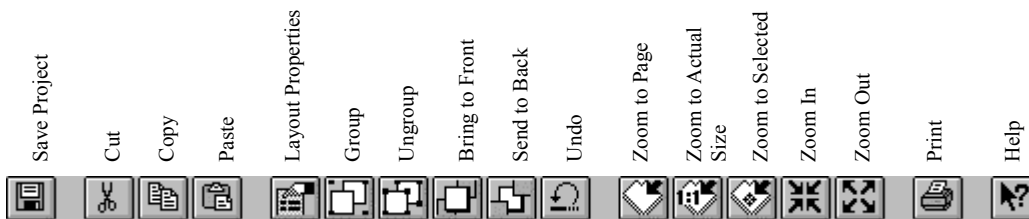


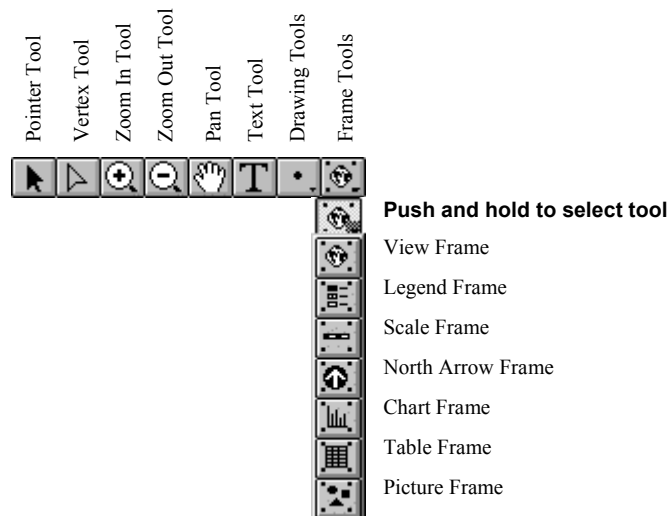
Chart - Tools



Layout - Buttons



Layout - Tools



For more Information: <http://www.esricanada.com>

Getting Started

Before you begin, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

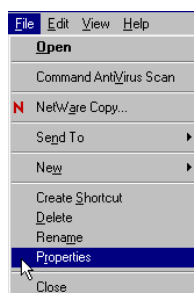
! (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /**.

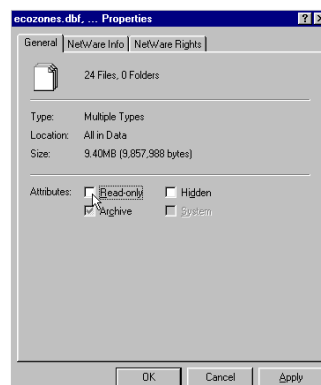
Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For windows based systems follow these instructions:

! (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

1. Open your new data folder
2. Choose **Edit > Select all** (they will all appear Blue)
3. From the menu choose:
File > Properties



4. Remove the check mark in the box beside the label **read only**.

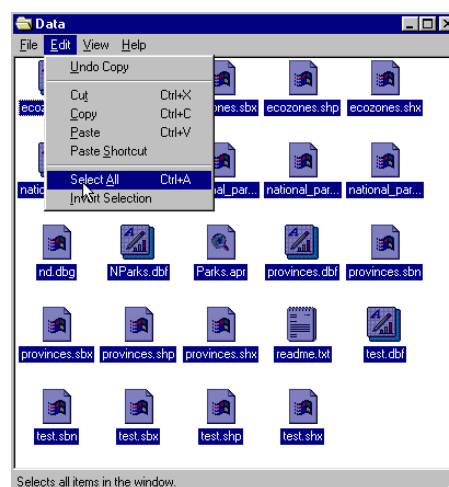


5. Click **apply**, then **OK**

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.



STARTING THE PROJECT

The purpose of this lesson is to include new data to a table by joining, display various themes by recoding polygons, displaying point data, building queries. We will look at geological age, rock type and mineral locations.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**



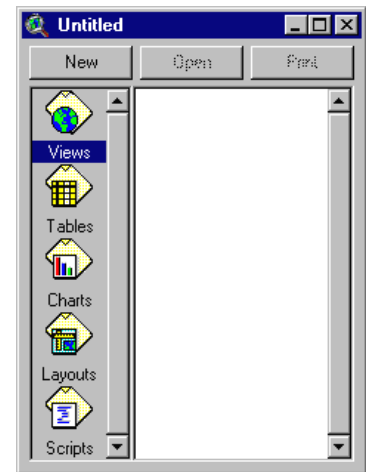
When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

! In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.

4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.

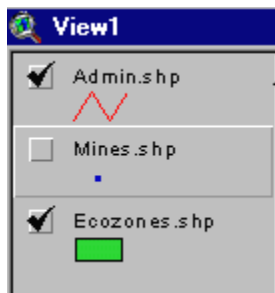


5. Click the **Add theme button**  and Navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
Ecozones.shp
mines.shp

6. Click **O.K.**

This will add these three themes to the table of contents of the new view. They are now available for display and editing.



7. **Click and drag** to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line themes should be on top of polygon files to be visible.

8. **Click** on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by **Clicking** on other themes.

9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **apr.** means that you are saving your work as a project that can be opened up in the future)

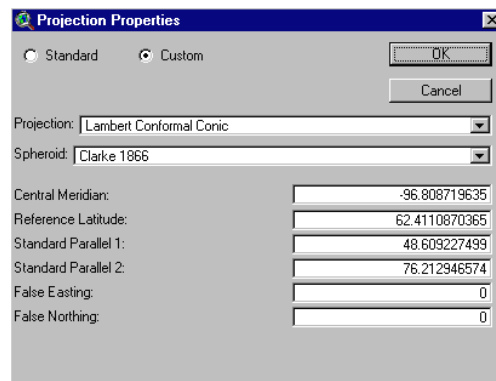
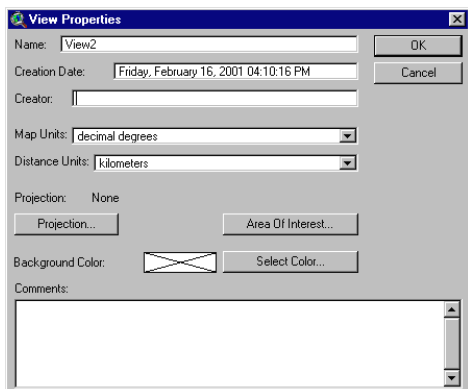
ArcView - Changing the Map Projection

! A **Map Projection** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

CHANGING PROJECTION

Now we will give our map “projection coordinates” to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. Change the **map units to decimal degrees**.
4. Select **Distance Units** list. Change the **distance units to kilometers**
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.

Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles).

Conformal implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.

Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The **central meridian**, **false easting** and **false northing** are used to define the X (easting) and Y (northing) coordinates of the map. The **reference latitude** and **standard parallel 1 and 2** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

8. Click **O.K.**

The bottom of the **View Properties** window should look like this:



9. Click **O.K.**

! Note the difference in the map presentation.

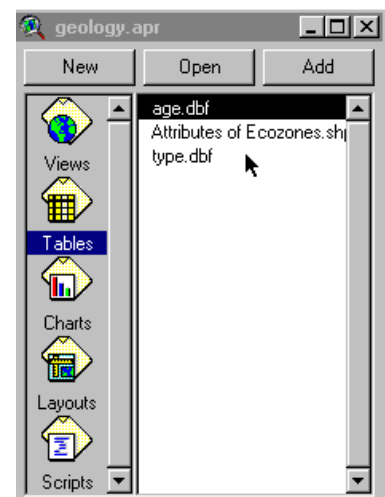
? How is it different from the non-referenced map?

ArcView - JOINING TABLES

1. Close the **View1** window.
2. From the **untitled project** window select **tables**.
3. Click the **Add button** and Navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

- **age.dbf**
- **ecozones.dbf**
- **type.dbf**

4. Select and Open each table.



? Examine each table and find a field common to all. What is it?

This is probably the field name containing the ecozones by name.

5. Select and activate the **age.dbf** table.
6. Click on the title of the field you identified as common to all tables.
7. From the **window menu**, select and activate the **ecozones.dbf** table.
8. Click on the title of the field you identified as common to all tables. It will appear darkened when selected.

! This common field will be used to join the two tables, with **age.dbf** being added to **ecozones.dbf**.

9. Click on the **Join** button .

? Examine the ecozones table.
What has been added at the end of the table?

Now Add rock type information.

10. Select and activate the **type.dbf** table using the **window menu**.
11. Click on the title of the field you identified as common to all tables.
12. From the **window menu**, Select and activate the **ecozones.dbf** table. Is the common field still selected? If not, click on it to make it active.
13. Click on the **Join** button. .

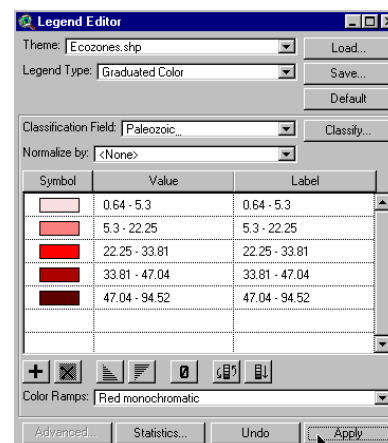
? Examine the attributes of ecozones table.
What has changed?

14. Close all the tables.

ArcView - DISPLAYING A MAP

Mapping Geological Age

1. From the **Project window** select the **Views** icon.
2. Select and open **View1**.
3. **Double Click** the **ecozones.shp** theme to edit it.
4. Open **Legend Type** list.
5. Select **Graduated Colour**.
6. Open **Classification Field** list.
7. Select **Precambria** (truncated to 10 characters from Precambrian)
8. Click **Apply**.
9. Close **Legend Editor** window.



! This will display the percent of Precambrian rock in each ecozone.

? Where are the oldest rocks located in Canada?

Describe the distribution of Precambrian rocks.

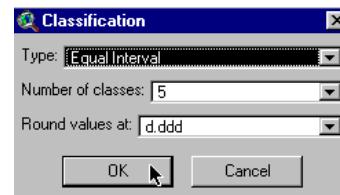
? Which ecozones have the greatest percent?

? How does this relate to the names of those ecozones?

Use the Identify Tool to  query an ecozone to get its name and other details.

ArcView - CLASSIFYING A MAP

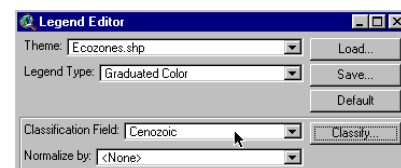
1. Double Click the **ecozone** theme.
2. Click **Classify**.
3. Select **Type Equal Intervals**.
4. Click **O.K.**
5. Click **Apply**.
6. Close **Legend Editor** window.



? How did the map and legend change?

Repeat this process to show the youngest rocks, (Cenozoic).

1. Double Click the **ecozone** theme.
2. Click **Classify**.
3. In the **Classification Field** list replace **Precambria** with **Cenozoic**.
4. Click **Apply**.
5. Close **Legend Editor** window.



? Where are the youngest rocks?

Describe the distribution of Cenozoic rocks.

Now, repeat this process again for both *Mesozoic* and *Paleozoic* rocks.

? Remember that the age of rock increases from the Cenozoic to the Mesozoic, to the Paleozoic and to Precambrian, the oldest.

Describe the distribution of geological rock age by ecozone.

Next, we'll map the ecozones by rock type.

1. Double Click the **ecozones.shp** theme to edit it.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.

5. Select **igneous**.
8. Click **Apply**.
9. Close **Legend Editor** window.

? Describe the distribution of igneous rocks.
Explain why they occur in these areas.

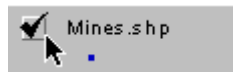
Repeat this process for the other rock types: metamorphic and sedimentary.

? Where do the metamorphic and sedimentary rock types appear?

ArcView - DISPLAYING POINT DATA

Let's look at the mineral deposits in Canada and see how they are associated with certain rock ages.

1. Check the **mines** theme to view it.



? What can you say about the distribution of mines in Canada?

? Which ecozones do not contain mines?

Canada mines both metallic and non-metallic elements. We will look at each of these classes and see where they occur.

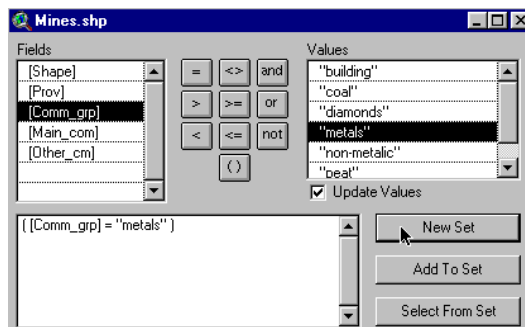
ArcView - BUILDING A QUERY

1. **Double Click** the **ecozones.shp** theme to edit it.
2. Open **Legend Type** list.
3. Select **Unique Value**.
4. Open **Classification Field** list.
5. Select **% prim_age**.
8. Click **Apply**.
9. Close **Legend Editor** window.

! This layer has the primary rock age for each ecozone.

Now let's build a query to find all the mines that extract metals.

1. Select and activate **mines** theme.
2. Click **Query Builder** button. 
3. **Double Click** Field **comm_grp**.
4. Click **=** button.
5. **Double Click** Values **metals**.
6. Click **New Set** button.
7. Close **query** window.



? With which age of rock is the majority of metal deposits associated?

8. Click **Clear Selected Features** button. 

Using the same process as used for *metals*, choose *building* (for building materials).

9. Click **Query Builder** button. 
10. **Double Click Field** *comm_grp*.
11. Click =.
12. **Double Click Values** *building*.
13. Click **New Set** button.
14. Close **query window**.

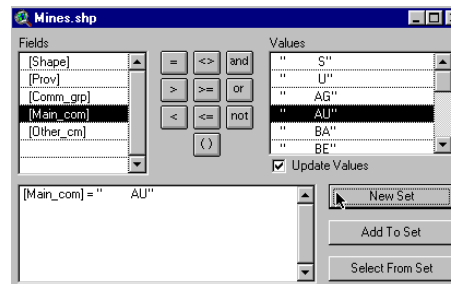
? From what rock age do most building materials come?

? What else does this map tell us about the location of mines and quarries for building materials?

Let's build a query to find where gold is most common.

Using the same process as for *metals* and *building*, substitute **Au** (Chemical symbol for gold) in the **values** field. Note that the **query builder / field** is now *main_com*.

1. Click **Query Builder** button.
2. **Double Click Field** *main_com*.
3. Click = button.
4. **Double Click Values** *Au*.
5. Click **New Set** button.
6. Close **query window**.



? Where do most gold mines appear?

? How many different groups of gold mines do you see?

Explore other types of minerals like Talc, Clay, Asbestos, Iron, Zinc and Salt.

? Are these minerals metallic or non-metallic?

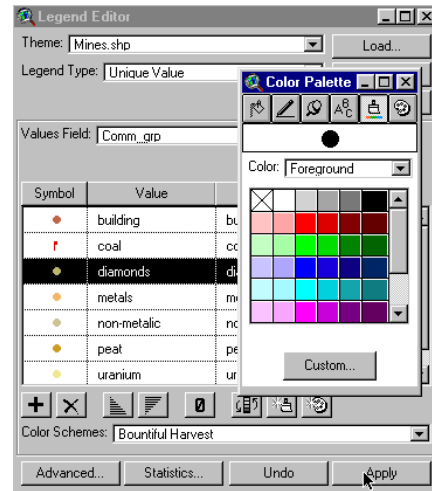
1. Keep your mine theme active.
2. From the **main menu**, select **Theme** and then **Table**.
3. You will find this information in the table you just open.

ArcView - CHANGING DISPLAY FEATURES

Now that we know how to display point data, let's try different symbols and colours for the points to make our map more meaningful.

1. **Double Click** the *mines* theme.
2. Open **Legend Type** list.
3. Select **Unique Value**.
4. Open **Values Field** list. Select *comm_grp*.
5. **Double Click** symbol next to *coal*.
6. Select a new symbol by clicking on it.
7. Click **Colour Brush** button.
8. Select a New Colour for the symbol representing coal.

! Note that the size of the symbol can be changed to suit your map scale.



? Prepare a layout for "Mines of Canada" by using the elements in *comm_grp* and displaying the various metals with different colours and symbols.

Or,

Prepare a map of Canada showing the relationship of selected types of mines and geological age or rock type by changing the map display.

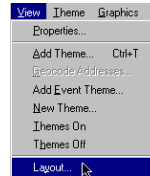
Try various combinations. Be creative!

ArcView - Map Layout for printing

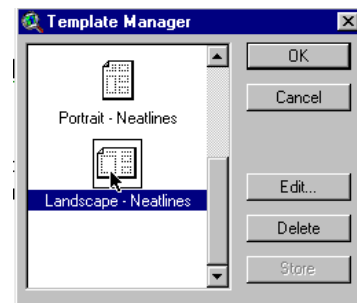
Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP

1. Click Menu **view**.
2. Click **Layout**.
3. Select the **layout template** you want to use (one of the **landscape** templates is suggested) .



4. Click **O.K.**
5. Maximize the **Layout** screen.



EDIT TITLE

1. **Double Click** on the title **View1**.
2. **Edit** the title to reflect the purpose of the map, e.g., forestry, population.
3. Click **O.K.**
4. **Resize** the title using the corner handles and move it into a position that is visually balanced.
5. **Click and Drag** to reposition the legend if necessary.

EDIT NORTH ARROW

1. **Double Click** on the *north arrow* and select a different presentation. Ensure that that the arrow is pointing in the representative north direction of the map.
2. **Click and Drag** to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. **Double Click** the *scale bar*.
2. **Choose** a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. **Change** the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing
5. **Click and Drag** to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. **Click** on the **Text** tool.
2. **Click** at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "produced by (your name, date)".
4. **Select** desired formatting information such as **alignment, spacing and rotation**.
5. **Click O.K.** when it appears, as you want it.
6. **Click** on **Pointer** tool.
7. **Select** the new text



8. **Click once** to change location or size, **Click twice** to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. **Click** menu **Layout**.
2. **Select Properties**.
3. **Change** the "**name**". Click **O.K.**

ARRANGING YOUR LAYOUT

- ! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)
- to **group** (tie pieces together so they move as one)
 - to **ungroup** (break apart the ties so that individual pieces move)
 - **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
 - **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
 - **zoom** (make image bigger or smaller)
 - **neat line** (border around layout)

PRINT MAP LAYOUT

1. **Click Print** button.
2. **Click Setup**.
3. **Select** your printer properties as outlined by your teacher.
4. **Click O.K.** to print.

! **Note** that the map can also be saved to a printer file.